PULLBACK AND PUSHFORWARD OF TILDE SHEAVES BETWEEN AFFINES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1475-1486, following the statement at 1458-1473.

Allowed standard prerequisites (editorial).

- The two tilde constructions agree. lemma-compare-constructions (card L26415, cited at ~1477). Identifies F_M with tilde(M) for part (1).
- Restriction of quasi-coherent sheaves to open Immersion-like morphisms. Modules, Lemma modules-lemma-restrict-quasi-coherent (cross-chapter, cited at ~1478). Engine for part (1).
- Affine schemes form a category equivalent to rings. lemma-category-affine-schemes (same chapter, cited at ~1466). Morphism-to-ring-map correspondence.

Source reference. Original lemma 01I2: sources/stacks-schemes/pinned-original-source.tex, source lines 1171–1171. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01I7: sources/stacks-schemes/pinned-original-source.tex, source lines 1295–1295. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01BJ: focused-seeds/geometry/S9-schemes-widetilde-pullback/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $(X, \mathcal{O}_X) = (\mathrm{Spec}(S), \mathcal{O}_{\mathrm{Spec}(S)})$, $(Y, \mathcal{O}_Y) = (\mathrm{Spec}(R), \mathcal{O}_{\mathrm{Spec}(R)})$ be affine schemes. Let $\psi : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of affine schemes, corresponding to the ring map $\psi^\sharp : R \to S$ (see Lemma 01I2).*

(1) *We have $\psi^*\widetilde{M} = \widetilde{S \otimes_R M}$ functorially in the R-module M.*
(2) *We have $\psi_*\widetilde{N} = \widetilde{N_R}$ functorially in the S-module N.*

Proof. The first assertion follows from the identification in Lemma 01I7 and the result of Modules, Lemma 01BJ. The second assertion follows from the fact that

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$\psi^{-1}(D(f)) = D(\psi^\sharp(f))$ and hence

$$\psi_*\widetilde{N}(D(f)) = \widetilde{N}(D(\psi^\sharp(f))) = N_{\psi^\sharp(f)} = (N_R)_f = \widetilde{N_R}(D(f))$$

as desired. $\square$